

PreFlow

Prepress Workflow System — Executive Overview

WS Display | Production Workflow Systems | September 2026

The Problem We're Solving

WS Display's prepress production workflow currently depends on three external systems: **Callas Toolbox** (for configuring processing recipes), the **Callas CLI servers** (for executing those recipes on every incoming job), and **NetSuite** as the ERP used internally to track jobs and surface file links. All three are being phased out.

PreFlow is the foundation of what replaces the Callas side of this — an internally built Python platform that gives WS Display full ownership of its prepress processing pipeline with no licensing dependency.

Current State (being replaced)

- Recipes configured in Callas Toolbox
- Artwork processed by Callas CLI servers (x2)
- Job links and notifications managed in NetSuite
- Dependent on licensed third-party software

Future State (where we're going)

- Recipes built and managed in PreFlow (Python)
- Artwork auto-processed by Python pipeline on EC2
- Job management handled by new custom ERP
- Internal platform — no Callas licenses required

What PreFlow Is

PreFlow is a browser-based tool for building, configuring, and validating the processing recipes that power the automated production pipeline. It was designed and is maintained by Brian Irick, WS Display's Production Workflow Systems Manager, as the control center for how artwork is processed — replacing the Callas Toolbox desktop environment.

A recipe is configured and validated in PreFlow, then exported to the production server. From that point the Python pipeline runs it automatically on every incoming job of that product type — no further intervention needed.

Configure recipes	Define preflight rules, color handling, imposition layouts, overlays, and cutpaths for
Validate & export	Confirm a recipe processes correctly, then push it to the production server — it runs automatically from the
Expandable by design	New preflight checks, processing steps, barcoding, sales order stamping, or any new capability can be add

The Automated Production Pipeline (Server Side)

Once recipes are deployed from PreFlow, the production pipeline runs fully automatically. No Callas. No manual steps in the prepress path:

Order received	Customer places order online and uploads artwork — stored in AWS S3
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Python pipeline runs	Artwork is preflight-checked, color-corrected, and imposed using the recipe deployed from PreFlow
Output to AWS	Proof JPEGs and press-ready PDFs are written back to S3
ERP records updated	New custom ERP receives S3 links, associated with the sales order
Customer notified	Customer is emailed a link to review and approve proofs on the WS Display website
Approved → print	Approved jobs route automatically to production — print files pulled from AWS and sent to press queues

Infrastructure: All assets — proofs, press files, overlays, cutpaths — hosted on **AWS S3**. Processing runs on **WS Display's EC2 server**. Python replaces Callas CLI — owned infrastructure, no licensing cost.

Getting PreFlow on Our EC2 Server

PreFlow deploys to WS Display's existing EC2 infrastructure with no code changes. Standing up PreFlow on EC2 is also the first step toward hosting the full Python processing pipeline on the same server — completing the Callas migration.

What's Needed	Detail
EC2 instance	Existing WS Display server — Ubuntu 22.04, t3.small or better
IAM role	S3 read/write access attached as a role — no hardcoded keys
Python stack	Python 3.11 + pip install -r requirements.txt
systemd service	Keeps app running; auto-restarts on reboot
Auto-deploy	Git pull + service restart triggered by GitHub push to main
Est. setup	2–4 hours for a developer familiar with EC2 and Linux